UNIT IV

17. (a) Find the inverse Laplace transform of $\frac{4s+5}{(s-1)^2(s+2)}$. (5)

(b) Using convolution theorem, find $L^{-1}\left[\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right]$. (6)

Or

18. (a) Solve : $y(x) = e^{-x} - 2\int_0^x y(u)\cos(x-u)du$ using Laplace transform. (5)

(b) Using Laplace transform solve $y''-3y'+2y = e^{-t}$ given $y(0)=1,\ y'(0)=0$. (6)

UNIT V

19. Find the Fourier transform of $f(x)$ if $f(x) = \begin{cases} 1-x^2 & if|x|\le 1 \\ 0 & if|x|>1 \end{cases}$. Hence prove that $\int_0^\infty \frac{\sin s - s\cos s}{s^3}\cos\left(\frac{s}{2}\right)ds = \frac{3\pi}{16}$. (11)

Or

20. Show that $e^{-x^2/2}$ is self reciprocal under Fourier cosine transform. (11)

5781007

B.Tech DEGREE EXAMINATION, NOVEMBER/DECEMBER 2019.

Second Semester

MATHEMATICS – II

Common to All Branches

(From 2013–14 Onwards)

Time : Three hours Maximum : 75 marks

SECTION A — ($10 \times 2 = 20$ marks)

Answer ALL the questions.

1. Find the eigen values of A^3 if $A = \begin{pmatrix} 3 & 4 & 1 \\ 0 & 4 & 5 \\ 0 & 0 & -1 \end{pmatrix}$

2. Find the quadratic form associated with the matrix $\begin{pmatrix} 5 & -7 \\ -7 & 2 \end{pmatrix}$

3. Calculate the *curl* of $\vec{F} = xyz\vec{i} + 3x^2y\vec{j} + (xz^2 - y^2z)\vec{k}$ and $(1,-1,1)$

4. If $\vec{F} = x^2\vec{i} + xy^2\vec{j}$ evaluate the line integral $\int_c \vec{F}.d\vec{r}$ from (0,0) to (1,1) along the path y = x.

5. Evaluate $L\left[e^{-2t}(t+\sin t)\right]$

6. Find $L\left[\frac{\sin at}{t}\right]$

7. Evaluate the inverse Laplace transform of $\frac{2s-1}{s^2+4s+5}$

8. Calculate: $\int_0^\infty e^{-t} t \sin t \, dt$.

9. Find the Fourier sine transform of $e^{-(5/2)x}$.

10. State Fourier integral theorem.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each Unit.

UNIT I

11. Reduce the quadratic form $2xy + 2xz + 2yz$ into canonical form by orthogonal reduction. Find also its rank, index, signature and nature. (11)

Or

12. Verify Cayley-Hamilton theorem for

$$A = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix}$$ Also find A^{-1}. (11)

UNIT II

13. (a) Find the angle between the surfaces $x \log z = y^2 - 1$ and $x^2 y = 2 - z$ at the point $(1,1,1)$. (5)

(b) Evaluate $\iint_S \vec{F}.\hat{n}\, dS$ where $\vec{F} = 4xz\vec{i} - y^2\vec{i} + yz\vec{k}$ and S is the surface of the cube bounded by $x = 0, x = a, y = 0, y = a, z = 0$ and $z = a$. (6)

Or

14. (a) Evaluate using Green's theorem in the plane for $\int_C (xy + y^2)dx + x^2 dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. (6)

(b) Prove that $\vec{f} = (y^2 \cos x + z^3)\vec{i} + (2y \sin x - 4)\vec{j} + 3xz^2\vec{k}$ is irrotational. Hence find its scalar potential.

UNIT III

15. (a) Verify the initial and final value theorem for the function $f(t) = 1 + e^{-t}(\sin t + \cos t)$ (6)

(b) Find the value $L\left[\frac{t \cos 2t}{e^{2t}}\right]$. (5)

Or

16. (a) Find the Laplace transform of the "square wave" function $f(t)$ is defined

$$f(t) = \begin{cases} k \text{ if } 0 \le t \le a \\ -k \text{ if } a < t \le 2a \end{cases}$$ and $f(t + 2a) = f(t)$

for all t. (6)

(b) Evaluate $L\left[\frac{\cos at - \cos bt}{t}\right]$, (5)